CBCT simulations in GATE (CNAO Half Fan HF data)

Overview

This pipeline generates synthetic CBCT images from CT data used as ground truth. Projection data are simulated in GATE using a Monte Carlo approach that models the system geometry, the X-ray source spectrum, and photon–matter interactions. The simulated projections are subsequently reconstructed into CBCT volumes using RTK. Although the method is analytical and therefore computationally demanding, it provides high accuracy and reliability, making the generated data suitable for validation studies.

Data

CT

The input CT must be non-normalized and include the background. Each CT volume must be converted to the CBCT coordinate convention using *align_flip.py*. For each dataset, the correct isocenter must be verified and specified in the conversion script. The isocenter can be typically found in the treatment plan. If this information is not available, it can be determined manually by visualizing the volume and identifying the center of tumor irradiation.

FILES MACRO (.mac)

GATE macros are ASCII text files (.mac extension) containing sequences of GEANT4 or GATE commands. Macros can be executed directly from the Linux command line or called from within other macros, enabling a modular and hierarchical simulation structure. A single macro, or a coordinated set of macros, must include all commands required to fully describe the simulation, following a logical execution order.

Macros typically define, in sequence:

- visualization settings
- geometry and volumes
- acquisition system
- detector configuration
- physics models
- simulation initialization
- source definition
- output management
- simulation execution

noscatterHF.mac

The noscatterHF.mac file simulates the **primary**, **attenuation**, and **flatfield** signals, providing ideal projection data in the absence of noise. It defines a detector with a resolution of 768 × 1024 pixels.

scatterHF.mac

The *scatterHF.mac* file simulates the **secondary** signal, which represents scatter, that is the contribution of secondary radiation reaching the detector and superimposing noise on the image. Since scatter is dominated by low spatial frequencies, the detector resolution in *scatterHF.mac* can be reduced to 96 × 128 pixels, corresponding to pixels 64 times larger. This increases the photon counts and gray-level values proportionally. The resulting gray levels must therefore be rescaled by dividing by 64.

Geometry files

Geometry files are provided in XML format and are used to import the specific geometry of the source-detector system (position and orientation)

The geometry file specifies:

- gantry and detector rotation angles in space,
- relative distances between the source, detector, and isocenter,
- detector translations along the reference axes,
- a 3 × 4 matrix defining the detector orientation with respect to the global reference coordinate system.

To simulate Half Fan (HF) projections at CNAO the geometry files are the followings:

- gGeometry_H1.xml : simulation geometry for *noscatter.mac* – detector-pose 1
- gGeometry_H2.xml : simulation geometry for *noscatter.mac* – detector-pose 2
- gGeometry_H1sub10.xml : simulation geometry for *scatter.mac* – detector-pose 1 (number of orientation subsampled of a factor 10)
- gGeometry_H2sub10.xml : simulation geometry for *scatter.mac* – detector-pose 2 (number of orientation subsampled of a factor 10)

NB: based on projection angles, the number of projections to be simulated is computed as:

N = (setTimeStop – setTimeStart) / setTimeSlice

[Here it's 501 *noscatterHF* simulations, and 51 *scatterHF* simulations].

GATE SIMULATION OUTPUT FILES

PRIMARY

These projections represent the contribution of primary (unscattered) photons emitted by the physical X-ray source and reaching the detector without undergoing any interaction in the traversed medium.

The scanned volume is treated purely from a geometrical perspective: it defines the photon trajectories but does not account for attenuation or scattering.
The signal includes the effects of the source spatial and angular distribution, collimation, geometric acceptance, and detector efficiency, but it does not include attenuation effects.

ATTENUATION

These projections represent the signal obtained from primary photons weighted by the attenuation of the traversed volume. The weighting factor is computed from the voxel-wise integral of linear attenuation coefficients. The contribution of scattered photons is not included. The projections therefore represent the effect of volume attenuation on the primary signal, assuming identical geometry and source distribution.

SECONDARY

These projections represent the signal generated by photons that have undergone at least one interaction within the volume, such as Compton or Rayleigh scattering. It corresponds to the scatter component of the signal, which acts as noise superimposed on the image. Events are weighted according to the interaction probabilities within the volume.

FLATFIELD

This projection represents the detector response in the absence of any attenuating object and under an ideal, spatially uniform source. The signal reflects only the intrinsic characteristics of the detector and collimation system. It is used for system normalization and correction of detector non-uniformities.

To summarize:

File	Cosa include	Cosa rappresenta
Primary	Unscattered primary photons	Ideal signal
Attenuation	Attenuated primary photons	Attenuated signal with no noise
Secondary	Scattered photons	Noise
Flatfield	Detector response to a uniform beam	Detector non-uniformities for normalization

From GATE documentation: One can separate compton, rayleigh and fluorescence photons, secondary (compton+rayleigh+fluorescence), primary or total (secondary+primary). flatfield is available to compute the measured primary signal if there is no object, which is useful for CT to apply the Beer Lambert law. ***The attenuation is ln(flatfield/primary) to get the line integral, i.e., the input of most CT reconstruction algorithms.***

➔ Since the attenuation files generated by GATE are computed as $ln\ (I_0/I_{\text{primary}})$ and do not include the contribution of scattered photons, additional processing is required to obtain realistic projection data. Specifically, the secondary signal is added to the primary signal, and the final attenuation is recomputed as

$ln\ (I_0/(I_{\text{primary}} + I_{\text{secondary}}))$ (check pipeline (old) step 4)

Data structure

The directory must be organized as follows.

The main folder (e.g. *simCBCT_HF_110*) contains the simulation setup and includes two subfolders: **data** and **mac**.

The **data** folder contains:

- CT volume
- Reference material database (Schneider2000MaterialsTable.txt)
- HU-to-density (g/cm^3) conversion table (Schneider2000DensitiesTable.txt)
- Geant4 material database with final composition and density (patient-HUmaterials.db)
- HU-to-material conversion map (patient-HU2mat.txt)
- RTK geometry files (for scatter and no-scatter simulations at the two detector positions)

The **mac** folder contains:

- noscatterHF_1.mac
- scatterHF_1.mac
- noscatterHF_2.mac
- scatterHF_2.mac
- 110kvp-3.2mmAl.mac – beam spectrum description
- verbose.mac

PIPELINE

Step 1

Convert the CT volume to the CBCT coordinate convention, ensuring that the correct isocenter is defined in the code.
Script to execute: "/mnt/ext1/fcasaccio/1_mha_data/ES/align_flip.py"

Step 2

Activate the conda environment **tf_15**.
From the main directory, execute the shell script: "/mnt/ext1/fcasaccio/1_mha_data/ES/simCBCT_pipeline.sh"

Step 3

Return to the **base** conda environment.
From the main directory, execute the shell script: "/mnt/ext1/fcasaccio/1_mha_data/ES/reconHF.sh"

Step 4

Average the two reconstructed volumes (*fdkBH_new1* and *fdkBH_new2*) to obtain the final attenuation volume. Convert to HU unit. Script "/mnt/ext1/fcasaccio/1_mha_data/ES /compute_final_vol.py"

The value for the water coefficient was derived empirically (check rescale_simCBCT.py).

PIPELINE (old – step by step) – for detector-pose 1 (same for 2)

Step 1

Convert the CT volume to the CBCT coordinate convention, ensuring that the correct isocenter is defined in the code. Script to execute: "*align_flip.py*"

Step 2

Activate the conda environment tf_15.
From the main directory, run the following commands:

gate_split_and_run.py mac/noscatterHF_1.mac -j 1

gate_split_and_run.py mac/scatterHF_1.mac -j 10 -o scatter1

The -j parameter specifies the number of jobs. For the no-scatter simulation, a single photon is simulated per job. For the scatter simulation, the computation is split into 10 independent jobs executed via SLURM.

Step 3

The outputs of the individual jobs are merged to obtain the final projection data by running the following command from the *scatter1* directory:

sbatch ../../gate_merge.slurm

Step 4

Execute the script "*processing1_HF.sh*").

For the secondary volumes (scatter simulations), the following operations are applied sequentially:

- Rescaling, to make the data comparable with the primary volumes by compensating for the different number of simulated photons (100,000 vs 1) and the lower detector resolution
- Resampling, to spatially align the secondary volumes with the primary data
- Interpolation, to achieve the same spatial resolution as the primary volumes

The processed secondary volumes are then added to the primary volumes to obtain final projections including scatter. The final projection is computed according to the Beer–Lambert law as

$$-\log\,(I/I_0),$$

where $I = I_{\text{primary}} + I_{\text{secondary}}$

and I_0 corresponds to the flatfield signal.

The results are saved in the *attenuation1* and *attenuation2* directories, corresponding to the two projection geometries.

Step 5

Run the script *reconHF.sh* in the base environment to reconstruct the final volumes from the projections for each projection geometry.

Step 6

Average the two reconstructed volumes to obtain the final simulated CBCT volume.